

ALPINE DROUGHT OBSERVATORY

Deliverable D.T2.3.1: Dataset Earth Observation

Prepared by Eurac Research

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Introduction

This report summarises the results of a study to assess the potential of different freely available soil moisture datasets to detect and quantify negative soil moisture anomalies. We considered four large scale and coarse resolution soil moisture datasets (ERA5, ERA5-Land, Lisflood, UERRA) and a local hydrological model (PREVAH), which acts as a benchmark. In-situ soil moisture from SwissSMEX, Wegenernet, Smosmania acted as a reference.

Datasets

The following section provides an overview of all datasets that went into this report.

Gridded soil moisture datasets datasets

- **ERA5:** [10.24381/cds.adbb2d47](https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-soil-moisture?tab=overview)
- **ERA5-Land:** [10.24381/cds.e2161bac](https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-land-soil-moisture?tab=overview)
- **UERRA (MESCAN-SURFEX):** [access through CDS](#)
- **LISFLOOD:** [10.23381/cds.3458969](https://cds.climate.copernicus.eu/cdsapp#!/dataset/lisflood-soil-moisture?tab=overview)
- **PREVAH** (Zappa and Gurtz, 2003; Viviroli *et al.*, 2009): PREVAH is a locally optimised hydrological model for Switzerland. It is used as an additional benchmark for the comparison with in-situ data from the SwissSMEX network.

TABLE 1: MAIN CHARACTERISTICS OF THE GRIDDED SOIL MOISTURE DATASETS.

DATASET-	SPATIAL RESOLUTION [KM]	TEMPORAL RESOLUTION-	TEMPORAL COVERAGE-
PREVAH	0.5	daily	1981-present
ERA5	~ 28	hourly	1950-present
ERA5-L	~ 9	hourly	1981-present
LISFLOOD	5	hourly	1991-present
UERRA	5.5	hourly	1961-2019

Surface Layer soil moisture on the 1 of June, 2018.

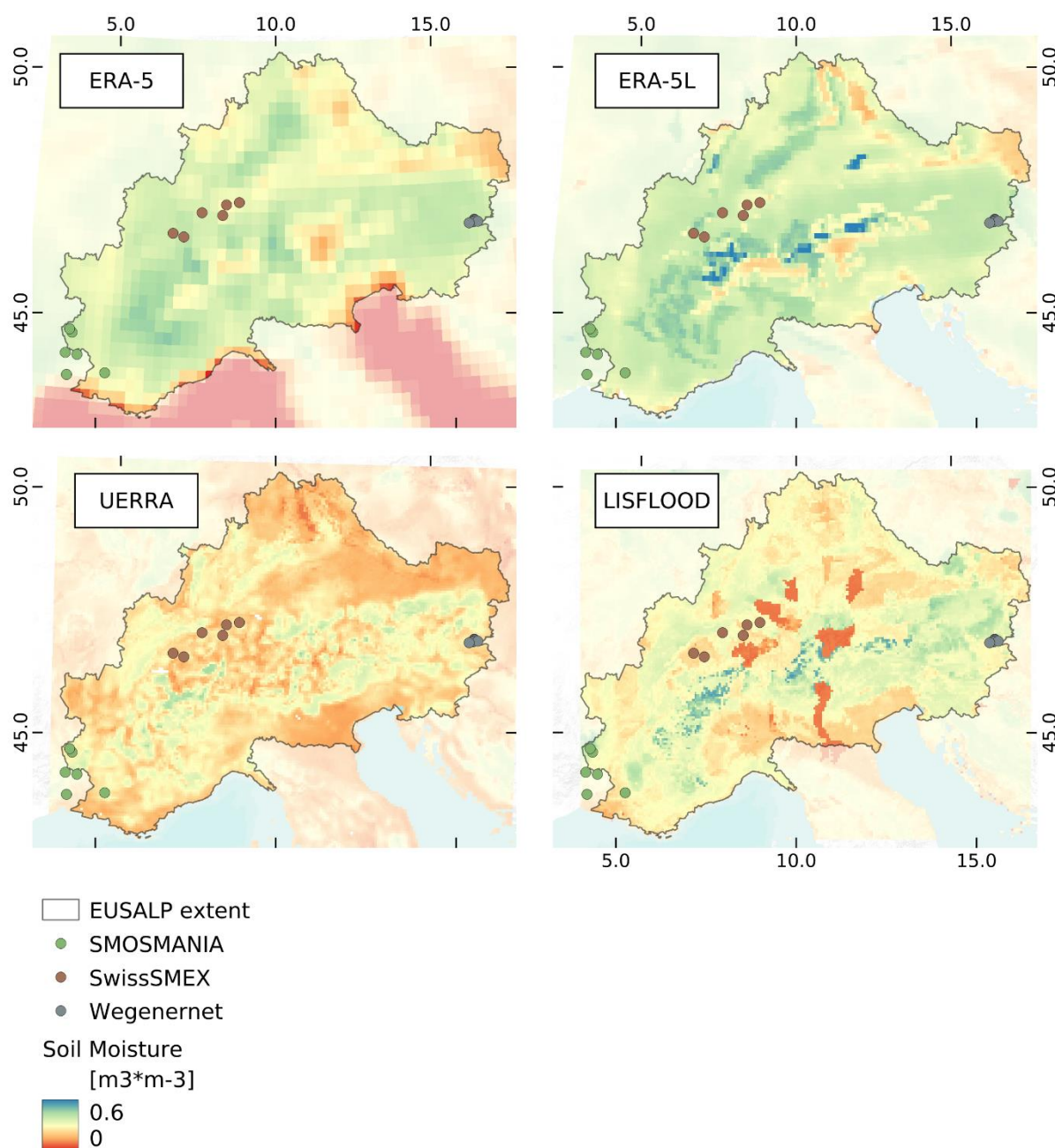


FIGURE 1: COMPARISON OF THE SPATIAL DISTRIBUTION OF SOIL MOISTURE ON THE 1ST OF JUNE, 2018.

The considered datasets have significantly different characteristics in terms of their spatial and temporal resolutions and coverages. Figure 1 shows example maps of the absolute soil moisture content for one day, the 1st of June, 2018. The gridded datasets differ in some of their key characteristics. Furthermore, ERA5, ERA5L, and LISFLOOD are updated operationally regularly, while UERRA is not.

Reference data

Data from the Swiss Soil Moisture Experiment (**SwissSMEX**) (Mittelbach and Seneviratne, 2012) provide measurements of the root zone soil moisture (50 cm) for six locations in Switzerland (see Figure 1). Stations from

two networks of the International Soil Moisture Network were used: six stations from the **SMOSAMANIA** network (Calvet *et al.*, 2007, 2016; Albergel *et al.*, 2008) provide measurements for the western Alps in France, and other seven stations of the **Wegenernet** (Kirchengast *et al.*, 2014; Fuchsberger, Kirchengast and Kabas, 2021) provide measurements for the Austrian Kalkalpen national park.

Methods

- SwissSMEX in-situ measurements were converted from mm to m³/m³: $SM_{vol} = SM_{mm}/500$
- The comparison is based on anomalies (SMA) to offset the impact of biases between the different models. We used the following equation to compute SMA for a given day t : $SMA_t = (SM_t - \overline{SM})/\sigma_{SM}$ where SM_t is the current value of SM, $\overline{SM}$ is the long-term average, and σ_{SM} the standard deviation. Each site's reference period was selected individually, based on the maximum temporal overlap between all models.
- Different datasets offer soil moisture representations at different soil depths. For the cross-comparison, values were interpolated to match the measured depths of in-situ datasets. (50 cm for SwissSMEX and 20 cm for SMOSMANIA and Wegenernet, respectively).
- To exclude the snow season from the analysis, we only considered the months April to September.

Results

We carried out a correlation analysis of the SMA time-series from each station for each network, both on the level of individual stations and the average conditions per network. The following table summarises the results of the correlation analysis in terms of the Pearson correlation coefficient (R). It presents the average R per soil moisture network:

TABLE 2: AVERAGE CORRELATION PER SOIL MOISTURE DATASET AND NETWORK

	SWISSMEX	SMOSMANIA	WEGENERNET
ERA5	0.785	0.501	0.591
ERA5-L	0.770	0.456	0.553
LISFLOOD	0.769	0.355	0.164
UERRA	0.778	0.618	0.589
PREVAH	0.835	-	

Table 3 presents the results of the SMA correlation analysis based on the average soil moisture conditions per network.

TABLE 3: SMA CORRELATIONS BASED ON AVERAGE SOIL MOISTURE CONDITIONS.

	SWISSMEX	SMOSMANIA	WEGENERNET
ERA5	0.615	0.822	0.750
ERA5-L	0.621	0.825	0.697
LISFLOOD	0.637	0.657	0.173
UERRA	0.660	0.884	0.734
PREVAH	0.644		

Figure 2 and Figure 3 show the time-series for 2012 and 2015 of SMA based on all considered model datasets compared to the SwissSMEX station Reckenholz.

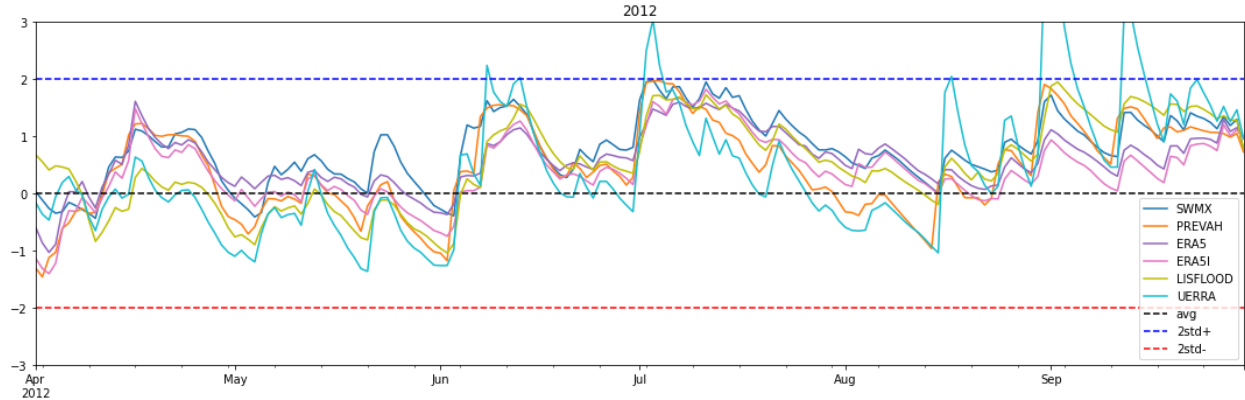


FIGURE 2: TIME-SERIES FOR 2012 OF THE SMAS BASED ON MODELLED SOIL MOISTURE COMPARED TO IN-SITU SMAS FROM SWISSMEX.

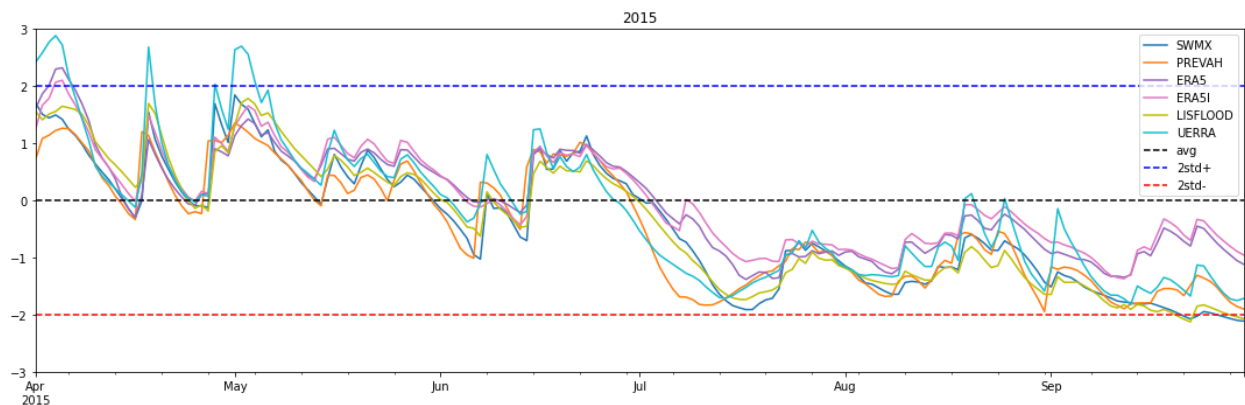


FIGURE 3: TIME-SERIES FOR 2015 OF THE SMAS BASED ON MODELLED SOIL MOISTURE COMPARED TO IN-SITU SMAS FROM SWISSMEX.

In general, the time-series of modelled SMA follow the temporal dynamics very well. In these two examples all models perform very similar. Figure 4 shows the scatterplots based on the complete based on the same time-series. There is a clear linear trend between modelled SMAs and measured SMA is evident. However, with Root-Mean-Squared-Errors between 0.45 and 0.65, the errors are relatively high. The two versions of ERA5 struggle to correctly quantify negative anomalies.

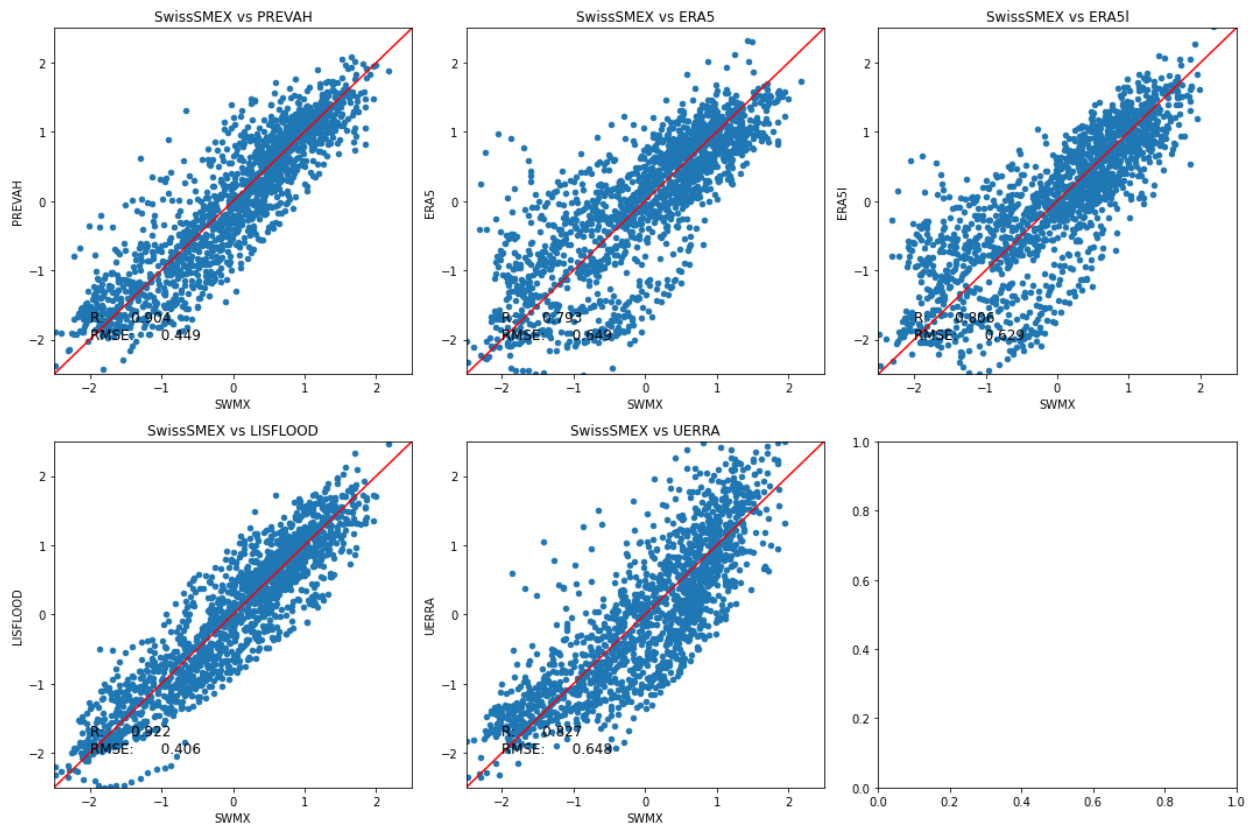


FIGURE 4: SCATTERPLOTS BETWEEN IN-SITU SMAS FROM SWISSMEX AND BASED ON THE FOUR DIFFERENT MODELS.

A look at the 2015 time-series (Figure 5) for the “Taenikon” station, another single station of the SwissSMEX network, gives an example for a case with systematic differences between the models. Here both versions of ERA5 estimate a return to normal conditions, after initially dry conditions in July. UERRA, LISFLOOD and the local PREVAH model correctly estimate the continued and worsened dry conditions.

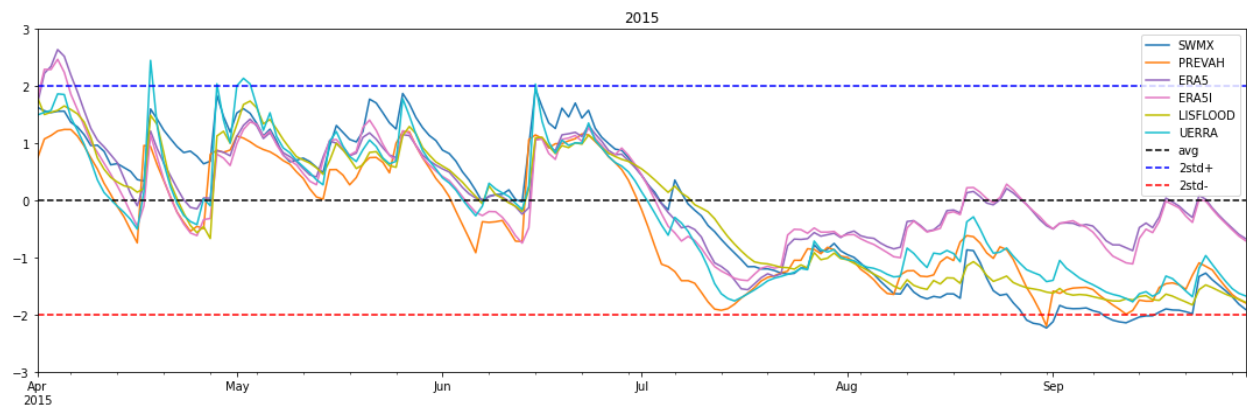


FIGURE 5: SMA TIME-SERIES FOR THE SWISSMEX STATION TAENIKON.

In another example for a single station of SMOSMANIA (Figure 6), we see how in some cases all datasets fail to correctly estimate SMA dynamics, and how they are affected by a significant degree of noise.

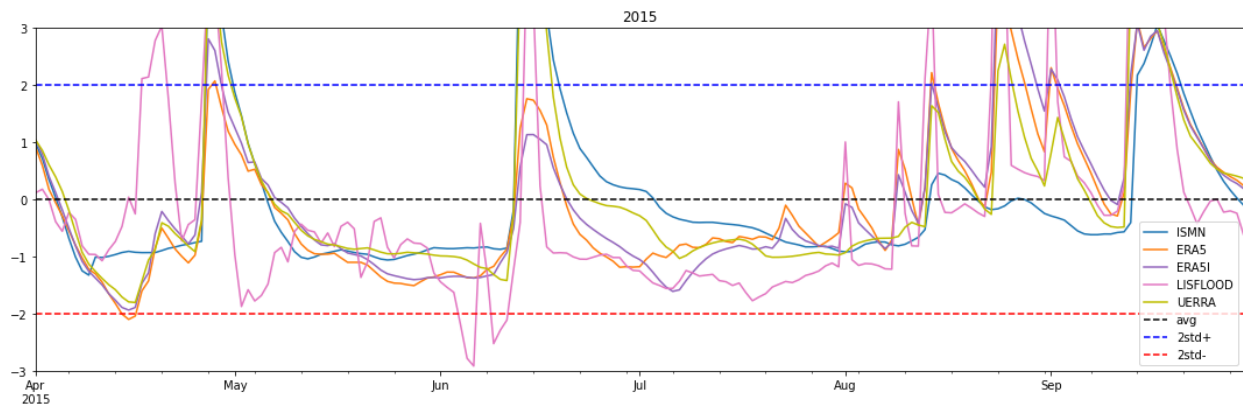


FIGURE 6: SMA TIME-SERIES FOR 2015, FOR STATION MEJANNES-LE-CLAP OF SMOSMANIA.

Observations and conclusions

- Based on the soil moisture conditions of single stations, SMAs based on ERA5 showed the best correlations (on average) with in-situ measurements.
- Higher spatial resolution does not lead to more accurate SMAs – ERA5 performs well even though it has, by far, the lowest spatial resolution.
- On average, all models can capture SMA dynamics similarly well.
- PREVAH, as a local hydrological model, outperforms ERA5, ERA5-L, LISFLOOD, and UERRA, but not by much.
- The results presented in Table 2 show significantly better correlations for SwissSMEX compared to the other in-situ networks.
- For Wegenernet, LISFLOOD performs significantly worse than the other models.
- Scatterplots show a clear linear relationship between in-situ measurements and modelled SMAs.
- The level of noise is high for all models, which hinders an accurate quantification of local anomalies.

Absolute accuracies differ between the different soil moisture networks. A more careful look at the uncertainties of in-situ measurements is required to draw final conclusions